Nuclear physics

30 questions: 15 multiple choice, 15 structured.

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1 · Part 1 Core The atom [1]

Which statement describes the structure of an atom?

A A negatively charged nucleus with positively charged electrons in orbit around it.

B A positively charged nucleus with negatively charged electrons in orbit around it.

C An uncharged nucleus with positively charged electrons in orbit around it.

D Positive and negative charges spread evenly through the whole atom, with no nucleus.

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2 · Part 1 Supplement The atom [1]

In a scattering experiment, a narrow beam of α-particles is directed at a very thin sheet of gold foil in a vacuum. Most of the α-particles pass straight through the foil with little or no deflection.

Which conclusion does this observation support?

A The nucleus is positively charged.

B The nucleus contains most of the mass of the atom.

C Electrons have a negative charge.

D Most of the atom is empty space.

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3 · Part 1 Core The nucleus [1]

A nucleus of cobalt is represented by $^{59}_27$ Co.

How many neutrons are in this nucleus?

A 32

B 27

C 59

D 86

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4 · Part 1 Core The nucleus [1]

Two atoms are isotopes of the same element.

Which quantity is the same for both of their nuclei?

A the number of neutrons

B the nucleon number

C the number of protons

D the mass of the nucleus

5 · Part 1 Supplement The nucleus [1]

Which statement describes nuclear fusion?

A A heavy nucleus splits into two lighter nuclei; the total mass decreases and energy is released.

B Two light nuclei join to form a heavier nucleus; the total mass decreases and energy is released.

C Two light nuclei join to form a heavier nucleus; the total mass increases and energy is released.

D Two light nuclei join to form a heavier nucleus; the total mass stays the same and energy is absorbed.

6 · Part 1 Supplement The nucleus [1]

An argon nucleus is $^{40}_{18}\text{Ar}$. A calcium nucleus is $^{40}_{20}\text{Ca}$.

Which row compares the relative charge and the relative mass of the two nuclei?

	relative charge	relative mass
A	the same	approximately the same
B	argon nucleus greater	approximately the same
C	the same	calcium nucleus greater
D	calcium nucleus greater	approximately the same

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7 · Part 1 Core Detection of radioactivity [1]

Which of these is **not** a significant source of background radiation?

A radio waves from a broadcasting mast

B radon gas in the air

C rocks and building materials

D cosmic rays

8 · Part 1 Core Detection of radioactivity [1]

A detector connected to a counter is placed near a radioactive source. The counter records 1440 counts in 4.0 minutes.

What is the average count rate in counts / s?

A 24

B 360

C 6.0

D 5760

9 · Part 1 Core The three types of nuclear emission [1]

Which row describes β-particles?

	nature	ionising effect	stopped by
A	fast-moving electrons	medium	a few millimetres of aluminium
B	fast-moving electrons	strong	a sheet of paper
C	helium nuclei	medium	a few millimetres of aluminium
D	electromagnetic radiation	weak	several centimetres of lead

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10 · Part 1 Supplement The three types of nuclear emission [1]

A narrow beam containing α-particles, β-particles and γ-radiation travels in a vacuum between two parallel metal plates. One plate is positively charged and the other plate is negatively charged. The beam enters midway between the plates, travelling parallel to them.

Which row describes what happens to each type of radiation?

	α-particles	**β-particles**	**γ-radiation**
A	large deflection towards the negative plate	small deflection towards the positive plate	not deflected
B	small deflection towards the negative plate	large deflection towards the positive plate	not deflected
C	small deflection towards the positive plate	large deflection towards the negative plate	not deflected
D	small deflection towards the negative plate	not deflected	large deflection towards the positive plate

11 · Part 1 Core Radioactive decay [1]

Which emissions from a nucleus cause the nucleus to change into a nucleus of a different element?

A γ-radiation only

B α-particles only

C α-particles, β-particles and γ-radiation

D α-particles and β-particles only

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12 · Part 1 Supplement Radioactive decay [1]

A nucleus emits one α-particle. The nucleus that is formed then emits two β-particles, one after the other.

How does the final nucleus compare with the original nucleus?

	proton number	nucleon number
A	2 less	4 less
B	4 less	4 less
C	the same	4 less
D	2 more	4 less

13 · Part 1 Core Half-life [1]

A radioactive sample gives a count rate of 720 counts / minute. The isotope in the sample has a half-life of 8.0 hours. Background radiation is negligible.

What is the count rate from the sample 24 hours later?

A 45 counts / minute

B 90 counts / minute

C 180 counts / minute

D 240 counts / minute

14 · Part 1 Core Safety precautions [1]

Which statement about the effect of ionising nuclear radiation on living things is correct?

A It can cause mutations, which may lead to cancer.

B It makes the cells radioactive, so that they emit radiation themselves.

C It can only harm cells when the radioactive source is inside the body.

D It has no effect on cells unless the dose is large enough to cause burns.

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15 · Part 1 Supplement Safety precautions [1]

A technician uses long-handled tongs to move a radioactive source.

How does using the tongs reduce the technician's exposure to radiation?

A The tongs absorb all of the radiation emitted by the source.

B The tongs make the source emit radiation more slowly.

C The tongs increase the distance between the source and the technician's body.

D The tongs reduce the half-life of the source.

16 · Part 1 Core The atom [5]

A neutral atom of an element has 11 protons in its nucleus.

(a) Describe the structure of an atom in terms of its nucleus and its electrons. [2]

(b) State the number of electrons in this neutral atom. [1]

(c) The atom loses one electron. State whether it becomes a positive ion or a negative ion. Explain your answer. [2]

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17 · Part 1 Core The nucleus [7]

Copper has two stable isotopes, $^{63}_{29}Cu$ and $^{65}_{29}Cu$.

(a) Define the term proton number. [1]

(b) Define the term nucleon number. [1]

(c) State the number of protons, the number of neutrons and the number of electrons in a neutral atom of $^{65}_{29}Cu$. [3]

(d) Explain why $^{63}_{29}Cu$ and $^{65}_{29}Cu$ are isotopes of the same element. [2]

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18 · Part 1 Supplement The nucleus [6]

The equation shows a nuclear reaction that takes place in a nuclear reactor.

$^{1}_{0}n + {}^{235}_{92}U \rightarrow {}^{140}_{54}Xe + {}^{94}_{Z}Sr + x\,{}^{1}_{0}n$

(a) Name this type of nuclear reaction. [1]

(b) Determine the value of Z. [1]

(c) Determine the value of x. Show your working. [2]

(d) Describe how the total mass of the particles after the reaction compares with the total mass of the particles before the reaction, and state what this means for the energy in the reaction. [2]

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19 · Part 1 Supplement The nucleus [4]

An iron nucleus is $^{56}_{26}$Fe. A nitrogen nucleus is $^{14}_{7}$N.

(a) State the relationship between the proton number of a nucleus and the relative charge on the nucleus. [1]

(b) State the relative charge on a $^{56}_{26}$Fe nucleus. [1]

(c) Determine how many times greater the relative mass of a $^{56}_{26}$Fe nucleus is than the relative mass of a $^{14}_{7}$N nucleus. Explain your answer. [2]

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20 · Part 1 Core Detection of radioactivity [4]

A detector placed in a room records a count rate even when no radioactive source is present. This is due to background radiation.

(a) State what is meant by background radiation. [1]

(b) Name three sources that make a significant contribution to background radiation. [3]

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21 · Part 1 Core Detection of radioactivity [5]

A student uses a detector connected to a counter to measure the background radiation in two places in a building. No radioactive source is present. The table shows the results.

place	number of counts	time / minutes
cellar	270	6.0
classroom	330	10.0

(a) Calculate the count rate in the cellar, in counts / minute. [2]

(b) Calculate the count rate in the classroom, in counts / minute. [1]

(c) The cellar is below ground level. Suggest one reason why the count rate in the cellar is higher than in the classroom. [1]

(d) The student repeats the measurement in the cellar for another 6.0 minutes. The number of counts is slightly different. Explain why. [1]

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22 · Part 1 Supplement Detection of radioactivity [6]

A scientist investigates a radioactive source in a laboratory.

With no source present, the counter records 150 counts in 5.0 minutes.

With the source placed near the detector, the counter records 2280 counts in 4.0 minutes.

(a) Calculate the background count rate, in counts / minute. [1]

(b) Calculate the count rate due to the source alone, in counts / minute. [3]

(c) The source is moved further from the detector. The measured count rate is now 45 counts / minute. Calculate the count rate due to the source alone. [1]

(d) Explain why, at this larger distance, it is especially important to correct for background radiation. [1]

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23 · Part 1 Core The three types of nuclear emission [5]

A student places different absorbers, one at a time, between a radioactive source and a detector connected to a counter. The detector is 2.0 cm from the source, and this distance is kept the same. Background radiation is negligible. The table shows the results.

absorber	count rate / counts per minute
none	820
sheet of paper	816
aluminium, 3 mm thick	210
aluminium, 6 mm thick	208

(a) Deduce which types of radiation the source emits. Give a reason for each of your decisions about α-particles, β-particles and γ-radiation. [3]

(b) State the nature of γ-radiation. [1]

(c) The emission of radiation from a nucleus is random in direction. State what this means. [1]

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24 · Part 1 Supplement The three types of nuclear emission [7]

(a) Explain, with reference to their kinetic energy and their electric charge, why α-particles are strongly ionising. [2]

(b) A narrow beam of β-particles travels horizontally from left to right in a vacuum. It passes through a uniform magnetic field that is directed out of the page. The β-particles are deflected upwards, towards the top of the page.

A beam of α-particles is then sent through the same magnetic field, also travelling from left to right.

(i) State the direction in which the α-particles are deflected. Explain your answer. [2]

(ii) State how the size of the deflection of the α-particles compares with that of the β-particles. Explain your answer. [2]

(c) A beam of γ-radiation is sent through the same magnetic field. State what happens to the γ-radiation and give a reason. [1]

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25 · Part 1 Core Radioactive decay [5]

(a) State what is meant by radioactive decay. [2]

(b) A nucleus emits an α-particle. State whether the nucleus that is formed is a nucleus of the same element or of a different element. Explain your answer. [2]

(c) A sample contains a very large number of nuclei of the same radioactive isotope. Explain why it is not possible to predict which nucleus will decay next. [1]

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26 · Part 1 Supplement Radioactive decay [5]

Phosphorus-32 is radioactive. It decays by β-emission to form an isotope of sulfur. The decay equation is

$^{32}_{15}P \rightarrow {}^{A}_{Z}S + {}^{0}_{-1}\beta$

(a) State the change that happens inside the nucleus during β-emission. [1]

(b) Determine the values of A and Z. [2]

(c) Some radioactive nuclei emit γ-radiation after an α-decay or β-decay. State the effect of emitting γ-radiation on the proton number and on the nucleon number of a nucleus. [1]

(d) A nucleus of $^{32}_{15}P$ has an excess of neutrons. Explain how β-emission makes the nucleus more stable. [1]

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27 · Part 1 Supplement Radioactive decay [6]

Radium-226, ^{226}_88 Ra, is a radioactive isotope. It decays by emitting an α-particle and forms a nucleus of radon (Rn).

(a) Suggest why a nucleus of ^{226}_88 Ra is unstable. [1]

(b) Write the decay equation for this α-decay, using nuclide notation for all three particles. [3]

(c) The radon nucleus formed is also radioactive. It decays by emitting another α-particle and forms a nucleus of polonium (Po). State the nucleon number and the proton number of this polonium nucleus. [2]

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28 · Part 1 Core Half-life [6]

A student measures the count rate from a sample of a radioactive isotope at regular intervals. Background radiation is negligible. The table shows the results.

time / hours	count rate / counts per minute
0	960
2.0	680
4.0	480
6.0	340
8.0	240
10.0	170

(a) Define the half-life of an isotope. [2]

(b) Use the table to determine the half-life of this isotope. Show how you used the table. [2]

(c) Predict the count rate from the sample at 16.0 hours. [2]

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29 · Part 1 Supplement Half-life [7]

A scientist measures the count rate near a radioactive sample. The readings have **not** been corrected for background radiation. The background count rate is 60 counts / minute.

time / minutes	measured count rate / counts per minute
0	700
5.0	513
10.0	380
15.0	286
20.0	220
25.0	173

(a) Calculate the count rate due to the sample alone at time 0. [1]

(b) Determine the half-life of the isotope in the sample. Show your working. [3]

(c) A factory makes paper. It uses a radioactive source on one side of the paper and a detector on the other side to monitor the thickness of the paper. The table shows four isotopes.

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isotope	radiation emitted	half-life
W	α	430 years
X	β	28 years
Y	γ	6.0 hours
Z	β	15 minutes

[3]

State which isotope is the most suitable. Explain your choice by referring to both the type of radiation and the half-life.

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30 · Part 1 Core Safety precautions [6]

A school keeps several small radioactive sources for use in physics lessons.

(a) State two effects of ionising nuclear radiation on living things. [2]

(b) Describe how the sources should be stored safely when they are not being used. [2]

(c) Describe two precautions that a teacher should take when moving and using a source during a lesson. [2]

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